Market Analysis: How Grocery Shoppers Buy

Customer purchasing behavior and product performance across 49,355 products and 63,100 user population

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Executive summary

Predictable habits, concentrated peaks and a loyal core of staple buyers

30,252
unique shoppers placed orders

10.53
items in the average basket

11.1 days
average gap between orders

~41%
of products (20,085) reordered more than once

When

Sunday and Monday carry 34.7% of orders; hourly demand peaks from 10 AM to 4 PM.

What

Fresh produce and dairy staples top both order counts and reorder rates.

So what

Subscriptions, mid-week deals and basket thresholds turn habits into growth.

Dataset at a glance

A relational MySQL schema (project_orders) analyzed in Power BI Desktop

49,355
products (SKUs)

134
aisles

21
departments

30,252
unique users

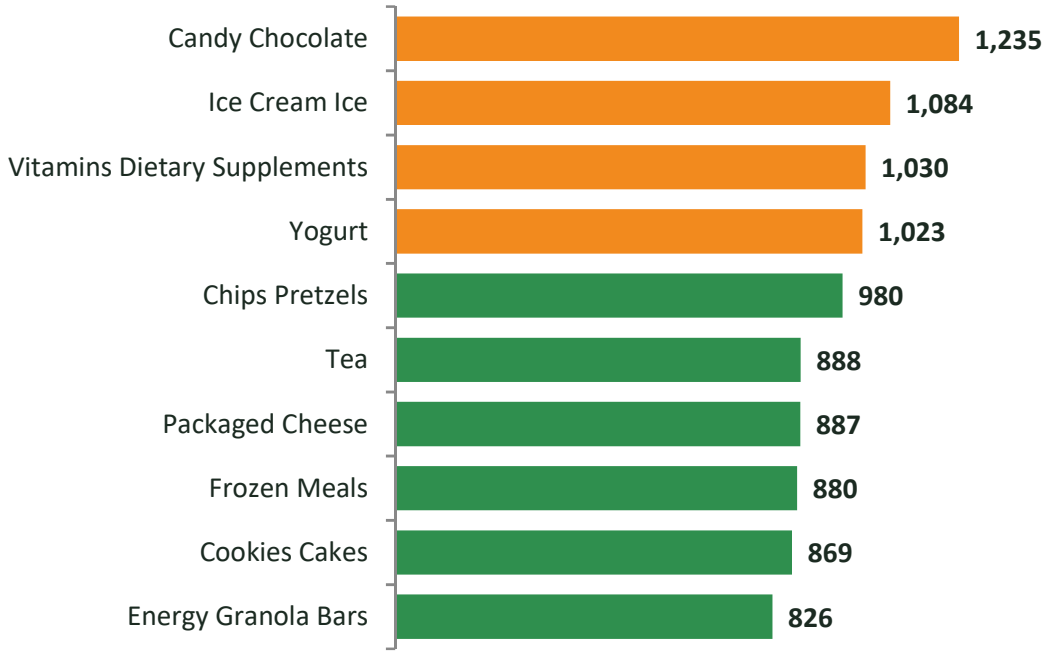
How the tables connect

products → aisles	aisle_id
products → departments	department_id
order_products_train → orders	order_id
order_products_train → products	product_id



Candy and snacks lead variety

Top 10 aisles by number of distinct products (SKUs)

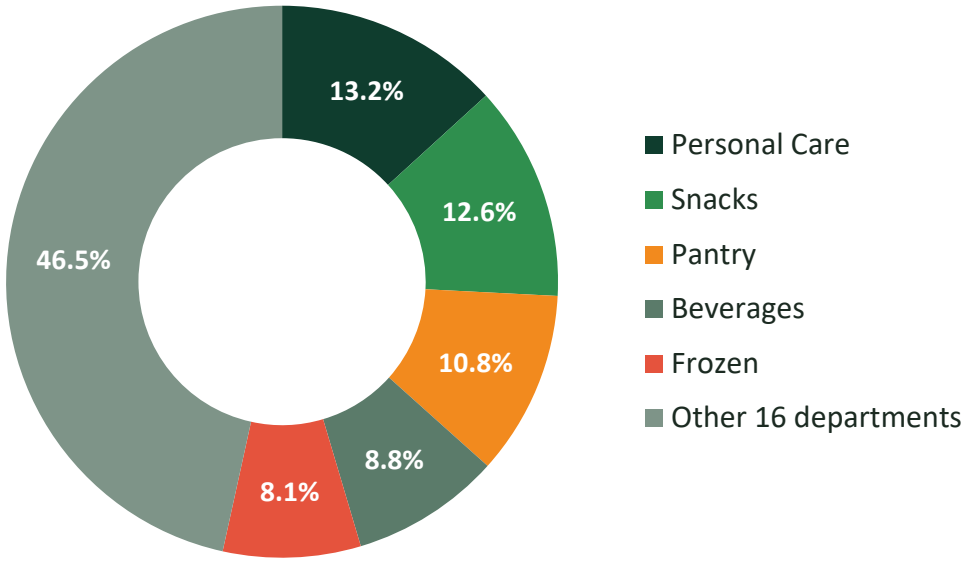


What it means

- Four aisles top 1,000 SKUs (highlighted)
- Impulse, snack and perishable aisles need the widest range
- Trim crowded ranges and add filters such as sugar-free, organic and vegan

Top 5 departments hold 53% of SKUs

Share of 49,355 catalog products by department (top 5 shown, rest grouped)

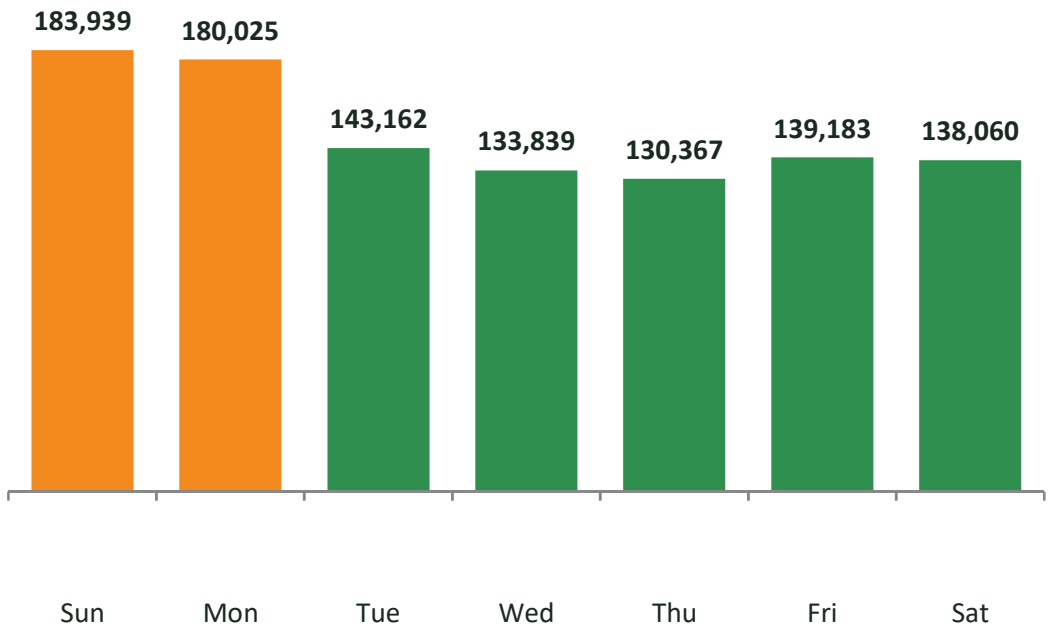


What it means

- Personal Care (13.2%), Snacks (12.6%) and Pantry (10.8%) are the broadest
- Deep ranges give choice but turn slower per SKU than Produce or Dairy Eggs
- Benchmark Snacks and Beverages suppliers for tiered volume discounts

Sunday and Monday dominate

Order volume by day of week



34.7%

of weekly orders land on Sunday and Monday

- Wednesday and Thursday run 27–29% below Sunday
- Run Mid-Week Flash Deals to even out warehouse and truck load

Orders peak from 10 AM to 4 PM

Hourly demand follows four clear phases

6–11 AM

Morning surge

9K to 88K orders per hour

11 AM–4 PM

Midday plateau

Above 82K orders per hour for 6 hours

5–9 PM

Evening taper

69K down to 24K orders per hour

1 AM–5 AM

Night lull

Under 4K orders per hour(lowest 1.6K at 3AM

Busiest hours (orders)

10 AM: 88,228 11 AM: 87,087
2 PM: 86,905 3 PM: 86,888

Act on it

Scale staff and servers 8:30 AM–5:30 PM; send app nudges at 9:30 AM and 3:30 PM

Weekly rhythm, and customers who lapse

Orders cluster at day 7, day 30 means 30+ days(uncapped).Remind just before day 7

11.1

days between orders on average (8.7 excluding the 30-day cap)

73%+

of active users reorder within 14 days

Weekly fresh reorderers

Day 7 spike. Offer fresh dairy and produce deals.

Lapsing Customers (30+ days)

Day 30 = 30 or more days (capped data). Offer win-back deals.

Trigger reorder reminders and cart-recovery emails on day 6 and day 11 after delivery, before shoppers defect.

Most baskets hold 5–14 items

Average basket is 10.53 items per order; very large baskets are rare

1-4 items	5-8 items	9-14 items	15-24 items	25+ items
23.2K	25.9K	26.1K	18.2K	6.2K
orders	orders	orders	orders	orders
Quick top-up runs	Core everyday baskets	The largest group	Above average stockers	Full-pantry trips are rare

Free-delivery threshold

A tiered minimum (around 10+ items) nudges 1–4 item runs upward.

Checkout cross-sell

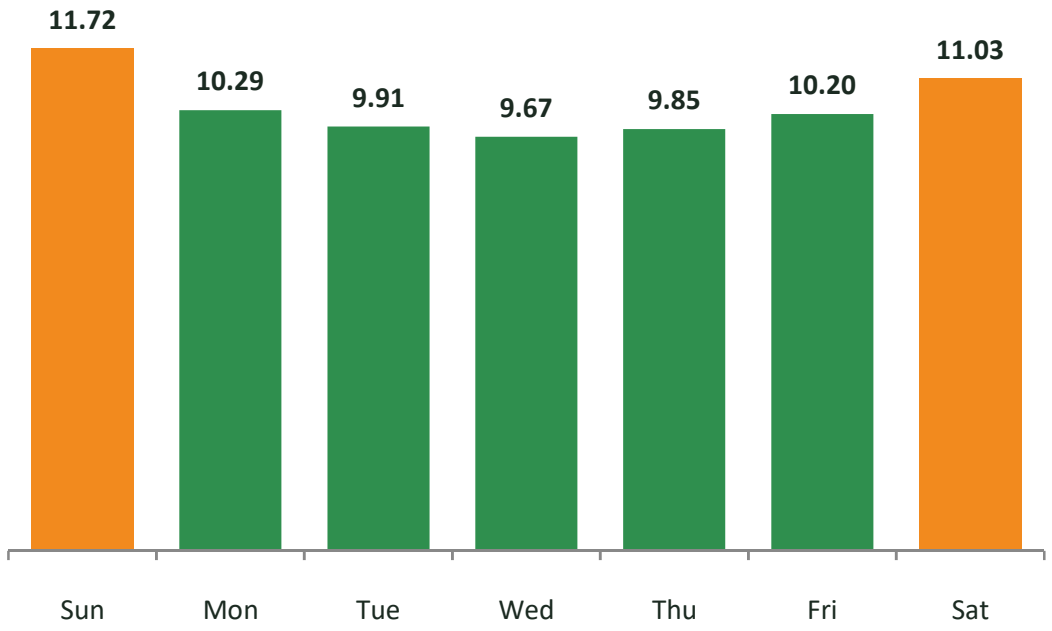
“Frequently Bought Together” moves 5–8 item carts into the 9–14 bracket.

Standard packaging

Orders cluster at 5–14 items, so standardize bags for 10–12 items.

Weekends carry bigger baskets

Average items per order by day of week



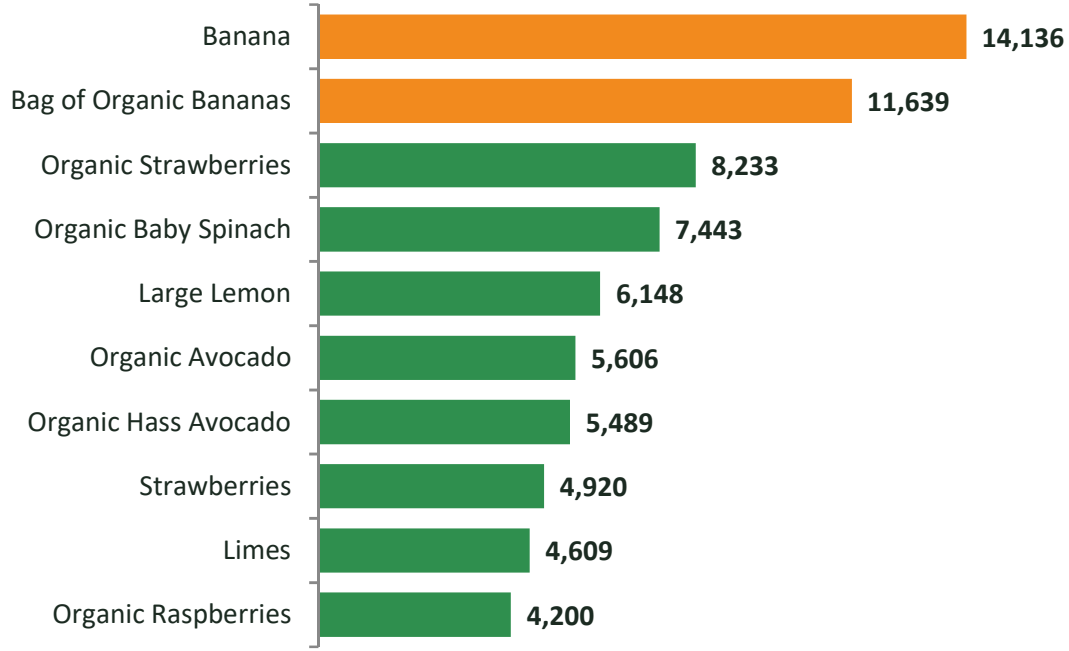
+21%

Sunday baskets (11.72 items) vs Wednesday (9.67)

- Weekend shoppers stock up for the week ahead
- Mid-week bundles like Buy 3, Get 1 on snacks and beverages can close the gap

Fresh produce fills the top 10

Total orders for the ten most-ordered products



What it means

- All ten are fresh fruit or vegetables
- Price these key value items sharply; shoppers judge the whole store by them
- Buffer cold-chain stock and diversify local growers

Milk and bananas win loyalty

Highest reorder rates among products with at least 100 purchases

#1

90.5%

Organic Low Fat Milk

258 of 285 orders were reorders

#2

88.5%

Organic Lowfat 1% Milk

330 of 373 orders were reorders

#3

88.2%

Banana

12,461 of 14,136 orders were reorders

Pattern

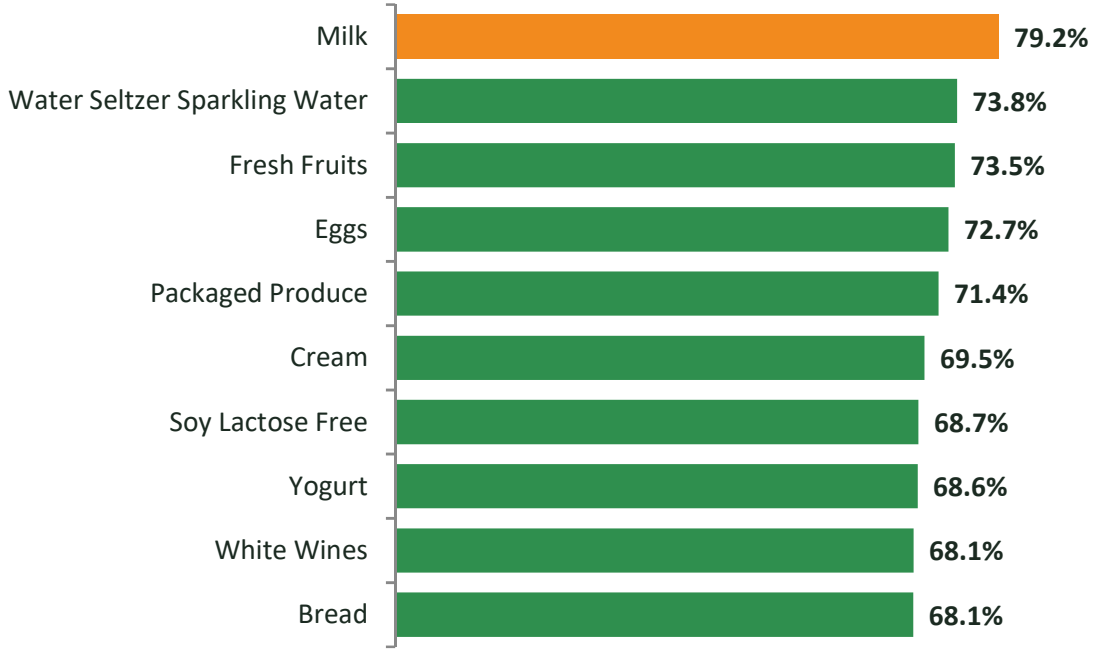
8 of the top 10 are milk variants, and all ten reorder above 86%.

Action

Offer Subscribe & Save with one-click reorder on every product above an 80% reorder rate.

Daily essentials reorder most

Average reorder rate by aisle, top 10



What it means

- Lowest: Spice seasonings 16.5%, Baking supplies décor 17.98%, Kitchen Supplies 18.02%
- Those are infrequent, durable purchases
- Lead the app home screen with high-reorder aisles and a Reorder All Weekly Essentials button

Every department has a flagship

Most reordered product by department (top 10 departments shown)

Department	Top reordered product	Reorders
Produce	Banana	12,461
Dairy Eggs	Organic Whole Milk	3,154
Beverages	Sparkling Water Grapefruit	2,027
Deli	Original Hummus	1,676
Bakery	100% Whole Wheat Bread	1,319
Frozen	Blueberries	1,051
Pantry	Extra Virgin Olive Oil	780
Canned Goods	Organic Black Beans	752
Household	100% Recycled Paper Towels	560
Breakfast	Honey Nut Cheerios	541



20,085

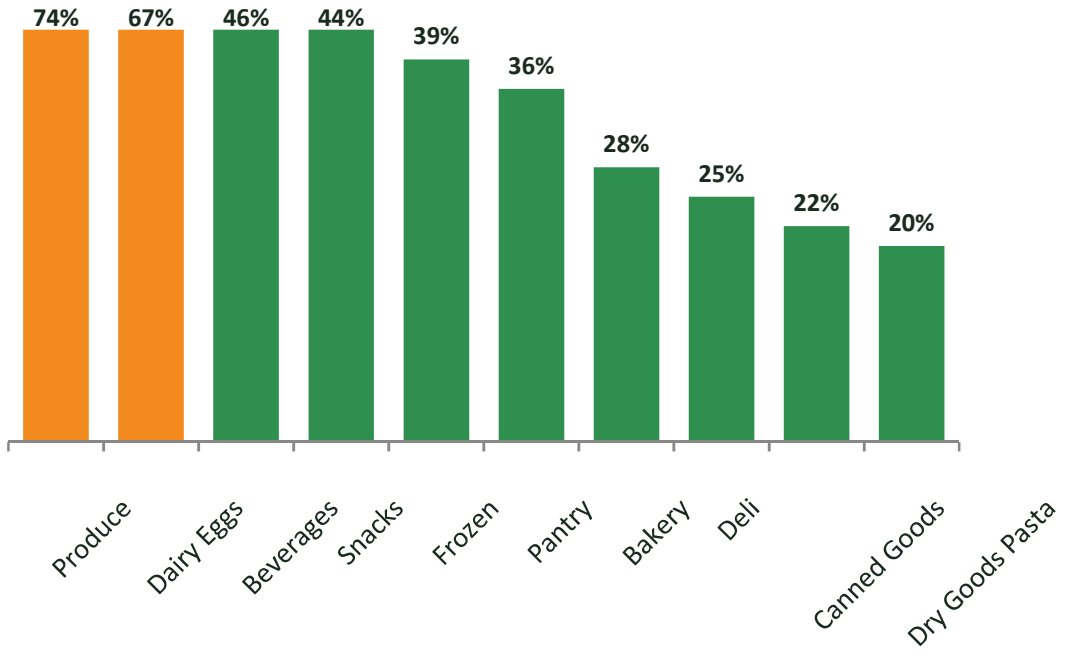
products reordered more than once (40.7% of catalog, 54.5% of the 36,864 ordered SKUs)

Items with 0–1 reorders are delisting candidates



Produce and dairy lead in reach

Share of 30,252 active users who ordered from each department

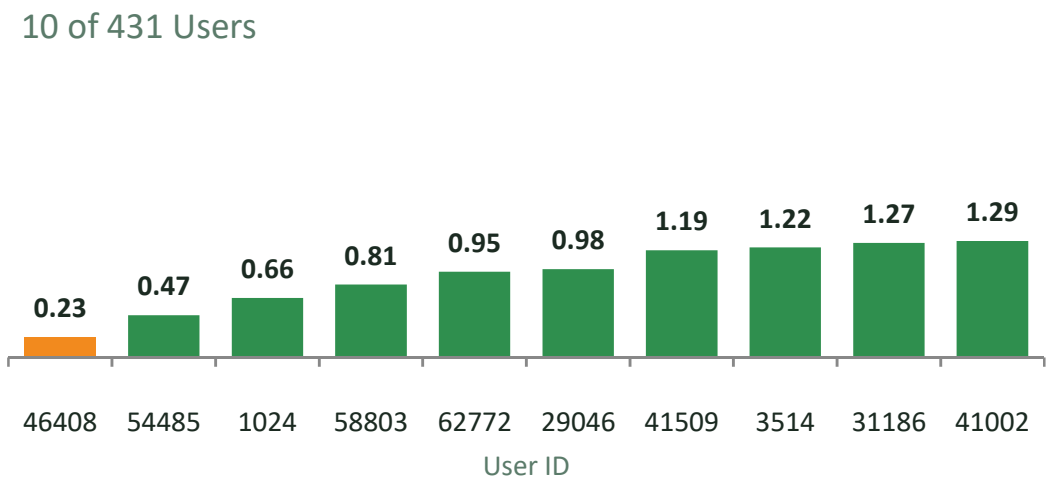


What it means

- Produce reaches 74% of users, Dairy Eggs 67%
- Alcohol (2.24%) and Bulk (1.07%) serve narrow niches
- Give shoppers who skip produce and dairy an intro voucher to build multi-department habits

Power users reorder almost daily

431 users hit the 100-order cap; ranked by fastest average gap



100

431 users hit the cap, mean gap 2.8 days

- Create a VIP tier with free priority delivery or auto-replenishment
- Lifetime-value models will undervalue them unless frequency and cart value are included

Five strategic moves

Each action is tied to a pattern in the data

1	**Fresh Pass subscription**	Auto-replenish staples with 80%+ reorder rates (milk, bananas) on a 7- or 14-day cycle.
2	**Mid-week pantry deals**	Discount non-perishables on Wednesday and Thursday to ease the Sunday and Monday peak.
3	**Basket thresholds**	Free delivery from about 10 items, plus Frequently Bought Together prompts at checkout.
4	**Produce price leadership**	Price bananas, avocados and berries sharply; produce reaches the most users.
5	**VIP loyalty tier**	Free priority slots for shoppers ordering every 2–4 days to protect the top decile.



Key takeaways

What the data says and where to act

Habits are predictable

Weekly cycles, Sunday and Monday peaks and a 10 AM–4 PM window make demand easy to plan for.

Staples build loyalty

Milk, bananas and other fresh essentials reorder at 86–91%, anchoring subscriptions and price perception.

Baskets are the lever

At 10.53 items per order, delivery thresholds and cross-sell prompts can lift order value.

Appendix: all 20 questions (1 of 3)

Each answer comes from the SQL and Power BI analysis; charts are on the slides noted

#	Question	Answer
1	Top 10 aisles by number of products	Candy Chocolate 1,235; Ice Cream Ice 1,084; Vitamins Dietary Supplements 1,030; Yogurt 1,023; Chips Pretzels 980 (full top 10 on slide 4)
2	Unique departments	21
3	Products across departments	Personal Care 6,524 (13.2%); Snacks 6,221 (12.6%); Pantry 5,331 (10.8%); Beverages 4,324; Frozen 3,982 (slide 5)
4	Top products by reorder rate	Organic Low Fat Milk 90.5%; Organic Lowfat 1% Milk 88.5%; Banana 88.2% (at least 100 purchases; slide 12)
5	Unique users who ordered	30,252
6	Average days between orders	11.1 days overall (8.7 excluding the 30-day cap); spike at day 7, day 30 means 30+ days; 431 users at the 100-order cap average 2.8 days
7	Peak hours of order placement	10 AM: 88,228; 11 AM: 87,087; 2 PM: 86,905; 3 PM: 86,888



Appendix: all 20 questions (2 of 3)

Each answer comes from the SQL and Power BI analysis; charts are on the slides noted

#	Question	Answer
8	Order volume by day of week	Sunday 183,939 and Monday 180,025 highest; Thursday lowest at 130,367 (slide 6)
9	Top 10 most ordered products	Banana 14,136; Bag of Organic Bananas 11,639; Organic Strawberries 8,233 (slide 11)
10	Users per department	Produce 22,303 (73.7%); Dairy Eggs 20,153 (66.6%); Beverages 14,024 (46.3%); Snacks 13,304 (43.9%)
11	Average products per order	10.53 items
12	Most reordered product per department	Produce: Banana 12,461; Dairy Eggs: Organic Whole Milk 3,154; Beverages: Sparkling Water Grapefruit 2,027 (slide 14)
13	Products reordered more than once	20,085 products (40.7% of catalog, 54.5% of the 36,864 ordered SKUs)
14	Average products added to cart per order	10.53 items



Appendix: all 20 questions (3 of 3)

Each answer comes from the SQL and Power BI analysis; charts are on the slides noted

#	Question	Answer
15	Orders by hour of day	About 9K at 6 AM, 88K at 10 AM, above 82K until 4 PM, 24K by 9 PM, 1.6–3.7K between 1 and 5 AM
16	Distribution of order sizes	1–4 items: 23.2K orders; 5–8: 25.9K; 9–14: 26.1K; 25+: 6.2K (slide 9)
17	Average reorder rate by aisle	Milk 79.2%; Water Seltzer Sparkling Water 73.8%; Fresh Fruits 73.5%; Eggs 72.7%; lowest: Spice seasonings 16.5%, (slide 13)
18	Average order size by weekday	Sunday 11.72 highest; Saturday 11.03; Wednesday 9.67 lowest (slide 10)
19	Top 10 users by number of orders	431 users tied at 100 orders (dataset cap); ranked by fastest gap, user 46408 orders every 0.23 days (slide 16)
20	Products per aisle and department	134 aisles roll up into 21 departments, for example Produce: fresh vegetables 564, fresh fruits 379; Snacks: Candy Chocolate 1,235

